



PlateLoop

Know what's eaten. **Waste less.**

An AI food scanner that measures what is served, eaten and left on every tray, so canteens cook the right amount, and everyone who eats gets their own nutrition feedback.

ECOLOOP HACKATHON · SINGAPORE

Avi · Akshaya · Disha · Ilseo
Nicole · Sophie
Class 3A15

1 The problem

755,000_t

of food was thrown away in Singapore in 2023. **Only about 18% was recycled.**

28%

of the world's food waste comes from **food service**: canteens, restaurants and caterers.

8–10%

of global greenhouse gas emissions come from **food that is lost or wasted.**



Canteens **cook by guesswork**, because nobody measures what people actually eat.

WL

"I get the same big scoop as everyone, and I'm told to finish it."

Wei Ling, 15, student

ML

"I cook by feel. Some days half the kailan comes back."

Mrs Lim, canteen operator

NA

"By the time we notice a patient isn't eating, it's been days."

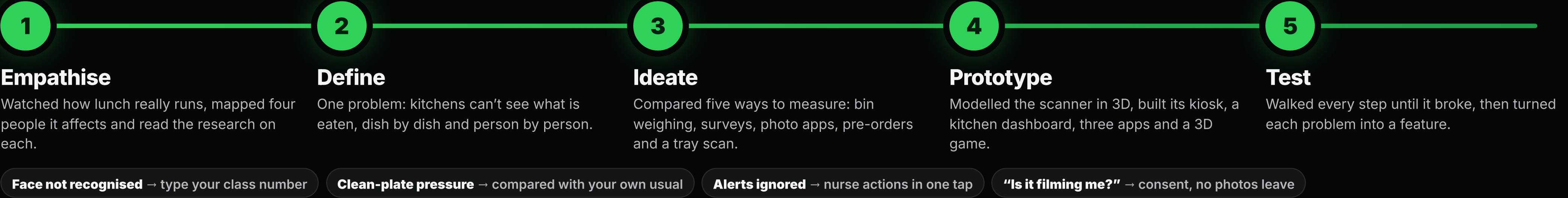
Nurse Aisha, Ward 7B

MT

"I want to build muscle, but I have no idea what the canteen gives me."

Marcus, office worker

2 Our process



Final prototype: PlateLoop Scanner

Look at the camera, put your tray down before you eat, and again after. The difference is exactly what you ate, dish by dish, in about two seconds.

620_g
SERVED

–

95_g
LEFT

=

525_g
EATEN

Sign in

Face match code, or type your number

Scan before

What you were served

Eat

Take what you'll finish

Scan after

Scraps weighed into the bin

- 1 Depth camera**

Finds each dish and measures its volume.
- 2 Face camera**

Signs you in with a code, never a stored photo.
- 3 10.1" screen**

Steps, your result and what you ate.
- 4 Weighing platform**

Checks the camera's estimate in real grams.
- 5 Food waste bin**

Weighs the scraps, so nothing goes uncounted.



3D model, rendered in Blender

3 One scan, four apps

each built from research on the people who use it

PlateLoop Kitchen CANTEN STAFF

- ✓ **Tomorrow's order** from real attendance, weather and events.
- ✓ **Waste by dish**, so the kailan nobody eats gets smaller.
- ✓ **Carbon ledger**: CO₂ avoided, ready for reporting.

\$S28,500

ingredients saved a year in an 800-meal school (illustrative)

Loopi STUDENTS

- ✓ A **virtual pet** fed by the lunch you really ate.
- ✓ **Food groups**, not calories.
- ✓ **Class races**, streaks and challenges.

–35%

plate waste when classes compete (schools review)

Loopi Care HOSPITALS

- ✓ **Every tray measured**, no food charts.
- ✓ Patients **choose tomorrow's meals**.
- ✓ Alerts reach the nurse **with the reason**.

29%

of inpatients were malnourished (Singapore study)

Loopi Work OFFICES

- ✓ A **daily pick** for your health goal.
- ✓ **Pre-order** and skip the queue.
- ✓ **Private**: teams, never names.

+51%

fruit ordered with pre-order nudges (canteen trial)

